

Nomad Connect 2023.1

Release Notes

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Features

Key	Summary:
RD-6999	The Dynamic Connection Manager (DCM) is a component in the Nomad Datacenters, which dynamically links trains to home agents using DNS. The CCUs send a DNS request to determine which HA (Home Agent) the CCU should connect to. DCM supports 2 major modes: - Fixed train assignment: the CCUs are linked to HAs in a 1-to-1 relationship using a configuration file. Updating the configuration file and running the DMC load-fixed-trains command updates the CCU-to-HA assignment in a few minutes for all trains at once. - Dynamic train assignment: the CCUs are linked to HAs dynamically. A list of CCUs is read from the NMS and the CCUs are spread evenly among the HAs. In this mode the HAs also have an availability state based on their state in the Zabbix datacenter monitoring. Both modes are configured on the component that runs in the CDC, which will always update DNS entries for the CCUs in a DNS service provider. The CCU will always bring up an Internet connection (e.g. cellular modem) to query its DNS entry and find its HA. When not connected, it will query more frequently, but when connected it will continue querying at a slower rate, so also already connected trains can be instructed to migrate to a different HA. For the CCU to switch to another HA, it is important that dynamic routing is configured or in the case of fixed train assignments, one could still rely on static routing and update the routing when changing the DCM fixed train assignment configuration.
RD-9846	Support was added to the engineering pages to allow for customer-specific branding and logos.
RD-9951	Support was added to the captive portal to allow for the dynamic modification of service class parameters.
RD-9952	Functionality was added to the passive scanning feature to allow for a list of hashed mac addresses seen during the scanning period to be included in the MQTT messages that are transmitted to the CDC.
RD-9979	Changes to the way the DANI OBN reporting module works to match on-board network design changes.
RD-9980	Changes to the way the DANI OBN reporting module works to match on-board network design changes.
RD-9995	Enhancements were made to the passive scanning feature to allow for the system to only send telemetry data when the train is actually moving above a certain configurable speed threshold.

<u>RD-10033</u>	The CCU-side wifi test tool support has been improved to avoid all disk reads as on some CCUs with slow (microSD) storage this was sometimes leading to bottlenecks.
<u>RD-10066</u>	The MQTT bridge has been enhanced to allow for a placeholder that holds the painted coach number which can in turn be configured and used in topics.
<u>RD-10111</u>	In a multi-ccu setup, the topics that Telemetry was using to publish the messages to were not using the correct coach and/or ccu number. This has been resolved.

Defects

Key	Summary:
<u>RD-10164</u>	A script that is used for Zabbix GPS monitoring purposes was sometimes returning a wrong value, causing wrong monitoring alerts. This has been resolved.
<u>RD-10132</u>	Due to an undocumented feature in the NROK/VTC APIs, the BIOS was also being updated when power cycling modems. This workaround ensures that modems are all enabled at boot time.
<u>RD-10104</u>	The SNR fields for modems were missing in the engineering pages, this has been resolved.
<u>RD-10082</u>	Critical Vulnerability: Debian DLA-3288-1 : curl - LTS security update
<u>RD-9990</u>	Critical Vulnerability: Debian DLA-3175-1 : python3.7 - LTS security update
<u>RD-9989</u>	Critical Vulnerability: Debian DLA-3204-1 : vim - LTS security update
<u>RD-9988</u>	Critical Vulnerability: Debian DLA-3243-1 : php7.3 - LTS security update
<u>RD-9987</u>	Critical Vulnerability: Debian DLA-3188-1 : sysstat - LTS security update
<u>RD-9738</u>	Under certain conditions, DCM was marking all HAs as unavailable. This has been resolved.
<u>RD-9611</u>	All java components have had their log4j versions upgraded to log4j version 2 in order to patch some critical vulnerabilities.
<u>RD-7577</u>	Due to the way the puppet manifests worked; it was not possible to remove crontabs that were no longer needed. The option to purge unwanted crontabs has been added.